🧭 8-Week ML + PyTorch Roadmap (Beginner, 1 hr/day)

Timeframe: Aug 11 – early Oct 2026 **Starting point:** Comfortable with Python + basic NumPy/Pandas. **Zero current ML/PyTorch knowledge. Time budget:** ~1 hr/day → roughly **50–55 total hours** before classes start.

A quick honest note first

At 1 hr/day starting from zero, you will **not** arrive at Brac as a multimodal/generative-AI expert — that's not realistic in 8 weeks, and it's not the goal. The real goal is: **walk in already comfortable with ML concepts and PyTorch, so your professor is teaching you research, not teaching you `for` loops in autograd.** Depth in multimodal/generative AI will come once you have an advisor and a specific research problem — this roadmap gets you fluent enough to keep up from day one, not to skip the learning that happens *in* the program.

Every week = ~7 hours. Each week has a small, concrete deliverable so you always know if you're on track.

Week 1: ML Foundations — What Are We Even Doing?

Goal: Understand what "machine learning" actually means before touching any library.

- [ ] What is supervised vs. unsupervised learning (with plain-language examples)
- [ ] What a "model," "features," "labels," "training," and "loss" mean
- [ ] Linear regression by hand — understand it as "fitting a line to minimize error"
- [ ] Gradient descent — the intuition (rolling downhill), not the full calculus derivation
- [ ] Train/test split, overfitting vs. underfitting (in plain terms, with a picture in your head)

Deliverable: In your own words (a short note to yourself), explain: what is a loss function, and why does the model try to minimize it?

Week 2: Classic ML with Scikit-Learn

Goal: Get hands-on with real (small) models before deep learning.

- [ ] Load a simple dataset (e.g. Titanic or Iris) with Pandas
- [ ] Train a Linear Regression and a Logistic Regression model in Scikit-Learn
- [ ] Train a simple Decision Tree / Random Forest classifier
- [ ] Evaluate with accuracy, precision/recall (just the definitions + when each matters)
- [ ] Plot results with Matplotlib (loss curve or confusion matrix)

Deliverable: One working notebook: load data → train 2 models → compare accuracy → one plot.

Week 3: PyTorch Basics — Tensors & Autograd

Goal: Learn PyTorch as a tool, from the ground up.

- [ ] What a tensor is (think: NumPy array that can track gradients)
- [ ] Basic tensor operations (you already know NumPy — map that knowledge over)
- [ ] `autograd`: what `.backward()` actually does, conceptually
- [ ] Build the **simplest possible neural network** by hand (a few `nn.Linear` layers)
- [ ] Write your first training loop: forward pass → loss → backward → optimizer step

Deliverable: A tiny PyTorch model trained on toy data (e.g. predicting y = 2x+1 from noisy points), fully understood line-by-line.

Week 4: Real Neural Networks — MLPs & CNNs

Goal: Go from toy examples to an actual image classifier.

- [ ] Build a Multi-Layer Perceptron (MLP) and train it on MNIST digits
- [ ] Learn what a Convolutional Neural Network (CNN) is and *why* it's used for images (intuition: local patterns, shared filters)
- [ ] Build and train a small CNN on MNIST or a small CIFAR-10 subset
- [ ] Learn the basics of activation functions (ReLU), and why we need them at all

Deliverable: A CNN that classifies MNIST digits, trained by you, with accuracy you can report.

Week 5: Meet the Transformer (Conceptually)

Goal: Understand *why* transformers matter — this is the backbone of everything modern (LLMs, VLMs, diffusion text-conditioning).

- [ ] What "attention" means, in plain language (a model deciding what to "look at")
- [ ] Why transformers replaced RNNs (parallelism, long-range context — at a conceptual level, not the full math)
- [ ] Use a pretrained model via Hugging Face's `pipeline()` (e.g. text classification or summarization) — just to see one work end-to-end
- [ ] Read a beginner-friendly explainer on "attention is all you need" (no need to derive the math yet)

Deliverable: Run 2–3 different Hugging Face pipelines (e.g. sentiment analysis, image classification) and write one sentence each on what's happening under the hood.

Week 6: First Look at Multimodal & Generative AI

Goal: Get oriented in your actual future research area — at a "know what it is" level, not "can build it" level yet.

- [ ] What CLIP does, in plain terms: it puts images and text into the same "space" so they can be compared

- [ ] Try a pretrained CLIP model via Hugging Face — zero-shot classify a few of your own images
- [ ] What a diffusion model does, in plain terms: it learns to turn noise into an image, step by step
- [ ] Run a pretrained text-to-image pipeline (e.g. via `diffusers`) and generate a few images — just to see the pipeline work

Deliverable: One short doc: 3 images classified with CLIP, 2–3 images generated with a diffusion pipeline, plus your own plain-English summary of what each model is doing.

Week 7: Small Guided Project

Goal: Put it all together in one small, complete project — your first "I built this" story.

- [ ] Pick ONE small project (don't overscope):
  - A CNN that classifies a small custom image dataset, or
  - A CLIP-based "search my photos by text description" tool
- [ ] Build it end-to-end: data → model/pipeline → results
- [ ] Keep notes on what worked, what didn't, and why (this is a research habit, start now)

Deliverable: One working mini-project in a GitHub repo with a short README.

Week 8: Consolidate & Get Research-Ready

Goal: Buffer week + basic research hygiene, so you walk in prepared, not just knowledgeable.

- [ ] Catch-up time for anything from Weeks 1–7 that didn't stick
- [ ] Learn basic Git/GitHub workflow if not already comfortable (commit, push, README)
- [ ] Skim (don't deep-read) 2–3 beginner-accessible blog posts or intros on multimodal/generative AI research to get familiar with the vocabulary your professors will use
- [ ] Review your Week 7 project and be ready to explain it out loud in 2 minutes

Deliverable: A short personal summary doc: what you learned, what you built, what you're still fuzzy on (bring this list to your first meeting with a professor/advisor).

🔗 Beginner-Friendly Resources

- **ML intuition:** StatQuest (YouTube) — best plain-language explanations of ML concepts
- **PyTorch:** "Learn PyTorch for Deep Learning" by Daniel Bourke (free, beginner-paced)
- **Transformers/Attention:** Jay Alammar's "The Illustrated Transformer" (visual, beginner-friendly)
- **Hugging Face:** the official `pipeline()` quickstart docs

- **CLIP/Diffusion intro:** Hugging Face's own beginner blog posts on CLIP and `diffusers`

How to use this realistically

- Missing a day is fine — don't try to "catch up" by cramming; just shift the week.
- If a week's material doesn't fully sink in, it's okay to spend 2 weeks on it and compress a later one — Week 8 is deliberately a buffer.
- The order matters more than the pace: don't jump to Week 6 (multimodal/diffusion) before Weeks 1–4 feel solid — it won't stick otherwise.